

CMSC216: Binary Encoding of Data

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Logistics

Reading

Bryant/O'Hallaron Ch 2.1-2.3

- ▶ Binary Encoding of Data
- ▶ Signed/Unsigned Integers
- ▶ Character Data
- ▶ Optional: Ch 2.4 Floats

Goals

- ▶ Wrap C discussion
- ▶ Integer and character encoding
- ▶ Negative Values and Overflow/Underflow
- ▶ Bitwise Operations, Byte-Ordering

Announcements

None

Unsigned Integers: Decimal and Binary

- ▶ Unsigned integers are always positive:
`unsigned int i = 12345;`
- ▶ To understand their binary encoding, first recall how decimal numbers “work” to encode quantities

Decimal: Base 10 Example

Each digit adds on a power 10

$80,345 = 5 \times 10^0 +$	5 ones
$4 \times 10^1 +$	40 tens
$3 \times 10^2 +$	300 hundreds
$0 \times 10^3 +$	0 thousands
8×10^4	80 tens of thousands
$5 + 40 + 300 + 80,000$	

Binary: Base 2 Example

Each digit adds on a power 2

$11001_2 = 1 \times 2^0 +$	1 ones
$0 \times 2^1 +$	0 twos
$0 \times 2^2 +$	0 fours
$1 \times 2^3 +$	8 eights
$1 \times 2^4 +$	16 sixteens
$= 1 + 8 + 16 = 25$	

So, $11001_2 = 25_{10}$

Optional Exercise: Convert Binary to Decimal

Base 2 Example:

$$\begin{aligned} 11001 &= 1 \times 2^0 + & 1 \\ &0 \times 2^1 + & 0 \\ &0 \times 2^2 + & 0 \\ &1 \times 2^3 + & 8 \\ &1 \times 2^4 + & 16 \\ &= 1 + 8 + 16 &= 25 \end{aligned}$$

So, $11001_2 = 25_{10}$

Try With a Neighbor

Convert the following two numbers from base 2 (binary) to base 10 (decimal)

► 111

► 11010

► 01100001

Answers: Convert Binary to Decimal

$$\begin{aligned}111_2 &= 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 \\&= 1 \times 4 + 1 \times 2 + 1 \times 1 \\&= 7_{10}\end{aligned}$$

$$\begin{aligned}11010_2 &= 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 \\&= 1 \times 16 + 1 \times 8 + 0 \times 4 + 1 \times 2 + 0 \times 1 \\&= 26_{10}\end{aligned}$$

$$\begin{aligned}01100001_2 &= 0 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 \\&\quad + 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 \\&= 0 \times 128 + 1 \times 64 + 1 \times 32 + 0 \times 16 \\&\quad + 0 \times 8 + 0 \times 4 + 0 \times 2 + 1 \times 1 \\&= 97_{10}\end{aligned}$$

Note: last example ignores leading 0's

The Other Direction: Decimal to Binary

Converting a number from base 10 to base 2 is easily done using repeated division by 2; keep track of **remainders**

Convert 124 to base 2:

$$124 \div 2 = 62 \text{ rem } 0$$

$$62 \div 2 = 31 \text{ rem } 0$$

$$31 \div 2 = 15 \text{ rem } 1$$

$$15 \div 2 = 7 \text{ rem } 1$$

$$7 \div 2 = 3 \text{ rem } 1$$

$$3 \div 2 = 1 \text{ rem } 1$$

$$1 \div 2 = 0 \text{ rem } 1$$

- ▶ Last step got 0 quotient so we're done.
- ▶ Binary digits are in **remainders in reverse**
- ▶ Answer: 1111100
- ▶ Check:

$$0 + 0 + 2^2 + 2^3 + 2^4 + 2^5 + 2^6 = 4 + 8 + 16 + 32 + 64 = 124$$

Decimal, Hexadecimal, Octal, Binary Notation

- ▶ Numbers exist independent of any writing system
- ▶ Can write the same number in a variety of bases
- ▶ C provides syntax for most common bases used in computing

	Decimal	Binary	Hex	Octal
Base	10	2	16	8
Mathematical	125	1111101 ₂	7D ₁₆	175 ₈
C Prefix	None	0b...	0x..	0...
C Example	125	0b1111101	0x7D	0175
printf()	"%d"	N/A	"%x"	"%o"

- ▶ **Hexadecimal** often used to express long-ish byte sequences
Larger than base 10 so for 10-15 uses letters A-F
- ▶ **Examine** `number_writing.c` and `table.c` for patterns
- ▶ **Expectation:** Gain familiarity with doing conversions between bases as it will be useful in practice

Hexadecimal: Base 16

- ▶ Hex: compact way to write bit sequences
- ▶ One byte is 8 bits
- ▶ Each Hex character represents 4 bits
- ▶ **Each Byte is 2 Hex Digits**

Byte	Hex	Dec
0101 0111	57 = $5 \cdot 16 + 7$	87
5 7		
0011 1100	3C = $3 \cdot 16 + 12$	60
3 C=12		
1110 0010	E2 = $14 \cdot 16 + 2$	226
E=14 2		

Hex to 4 bit equivalence

Dec	Bits	Hex
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Exercise: Conversion Tricks for Hex and Octal

How do bits naturally group in Hex and Octal?

Decimal	Byte = 8bits	Byte by 4	Hexadecimal
87	01010111	bin: 0101 0111 hex: 5 7	57 = 5*16 + 7 hex dec
60	00111100	bin: 0011 1100 hex: 3 C=12	3C = 3*16 + 12 hex dec
226	11100010	bin: 1110 0010 hex: E=14 2	E2 = 14*16 + 2 hex dec
Decimal	Byte = 8bits	Byte by 3	Octal
87	01010111	bin: 01 010 111 oct: 1 2 7	127 = 1*8 ² + 2*8 + 7 oct dec
60	00111100	bin: 00 111 100 oct: 0 7 4	074 = 0*8 ² + 7*8 + 4 oct dec
226	11100010	bin: 11 100 010 oct: 3 4 2	342 = 3*8 ² + 4*8 + 2 oct dec

Answers: Conversion Tricks for Hex and Octal

- ▶ Converting between Binary and Hexadecimal is easiest when grouping bits by 4: each 4 bits corresponds to one hexadecimal digit

bin: 0101 0111	bin: 1110 0010
hex: 5 7	hex: E=14 2

- ▶ Converting between Binary and Octal is easiest when grouping bits by 3: each 3 bits corresponds to one octal digit

bin: 01 010 111	bin: 11 100 010
oct: 1 2 7	oct: 3 4 2

Character Coding Conventions

- ▶ Would be hard for people to share words if they interpreted bits as letters differently
- ▶ **ASCII**: American Standard Code for Information Interchange
An old standard for bit/character correspondence
- ▶ 7 bits per character, includes upper, lower case, punctuation

Dec	Hex	Binary	Char	Dec	Hex	Binary	Char
65	41	01000001	A	78	4E	01001110	N
66	42	01000010	B	79	4F	01001111	O
67	43	01000011	C	80	50	01010000	P
68	44	01000100	D	81	51	01010001	Q
69	45	01000101	E	82	52	01010010	R
70	46	01000110	F	83	53	01010011	S
71	47	01000111	G	84	54	01010100	T
72	48	01001000	H	85	55	01010101	U
73	49	01001001	I	86	56	01010110	V
74	4A	01001010	J	87	57	01010111	W
75	4B	01001011	K	88	58	01011000	X
76	4C	01001100	L	89	59	01011001	Y
77	4D	01001101	M	90	5A	01011010	Z
91	5B	01011101	[	97	61	01100001	a
92	5C	01011110	\	98	62	01100010	b

Partial Table of ASCII Codes / Values, try `man 7 ascii` in a terminal for a full table

Exercise: Characters vs Numbers

Explain the following program and its output

```
1 // char_ints.c:
2 #include <stdio.h>
3 #include <string.h>
4 int main(){
5     ...
6     char nums[64] = {
7         72, 101, 108, 108, 111, 32,
8         87, 111, 114, 108, 100, 33,
9         0
10    };
11    printf("%s\n",nums);
12    len = strlen(nums);
13    for(int i=0; i<len; i++){
14        printf("[%2d] %c %3d %02X\n",
15                i,nums[i],nums[i],nums[i]);
16    }
17    return 0;
18 }
```

```
>> gcc char_ints.c
>> ./a.out
...
Hello World!
[ 0] H  72 48
[ 1] e 101 65
[ 2] l 108 6C
[ 3] l 108 6C
[ 4] o 111 6F
[ 5]   32 20
[ 6] W  87 57
[ 7] o 111 6F
[ 8] r 114 72
[ 9] l 108 6C
[10] d 100 64
[11] !  33 21
```

Answers: Characters vs Numbers

The Whole Array

```
char nums[64] = {  
    72, 101, 108, 108, 111, 32,  
    87, 111, 114, 108, 100, 33,  
    0  
};
```

Lays out a bit pattern at each spot the array; bit pattern is specified with decimal numbers

```
printf("%s\n",nums);
```

Print the array as though it were "string": an array of characters that is null terminated

Elements of the Array

```
printf("[%2d] %c %3d %02X\n",  
        i,nums[i],nums[i],nums[i]);
```

Print a single element of the array as

- ▶ %c : a character (ASCII table lookup for the glyph to draw)
- ▶ %3d : a decimal number (padding to width 3)
- ▶ %02X : as a hexadecimal number (with leading 0's if needed and padded with width 2 - *noice!*)

Unicode

- ▶ *World:* Why can't I write
컴퓨터
in my code/web address/email?
- ▶ *America:* ASCII has 128 chars.
Deal with it.
- ▶ *World:* Seriously?
- ▶ *America:* We invented
computers. 'Merica!



- ▶ *World:*
- ▶ *America:* ... Unicode?
- ▶ *World:* But my language takes
more bytes than American.
- ▶ *America:* Deal with it. 'Merica!

- ▶ ASCII Uses 7 bits per char,
limited to 128 characters
- ▶ UTF-8 uses **1-4 bytes per character** to represent **many**
more characters
(1,112,064 *codepoints*)
- ▶ Uses 8th bit in a byte
(**high-order bit**) to indicate
extension to multiple bytes
- ▶ Requires software to understand
coding convention allowing
broader language support
- ▶ ASCII is a proper subset of
UTF-8 making UTF-8
backwards compatible and
wildly popular

Binary Integer Addition/Subtraction

Adding/subtracting in binary works the same as with decimal
EXCEPT that carries occur on values of 2 rather than 10

ADDITION #1

```
  1 11      <-carries
  0100 1010 = 74
+ 0101 1001 = 89
-----
  1010 0011 = 163
```

ADDITION #2

```
  1111   1 <-carries
  0110 1101 = 109
+ 0111 1001 = 121
-----
  1110 0110 = 230
```

SUBTRACTION #1

```
          ? <-carries
  0111 1001 = 121
- 0001 0011 =  19
-----
```

```
VVVVVVVVVVVVVV
VVVVVVVVVVVVVV
VVVVVVVVVVVVVV
```

```
          x12 <-carries
  0111 0001 = 119
- 0001 0011 =  19
-----
  0110 0110 = 102
```

When 0/1 is represented as Low/High Voltage, one can design
digital circuits that implement arithmetic

Two's Complement Integers: Representing Negative Values

- ▶ To represent negative integers, must choose a **different** coding system than for positive-only integers
- ▶ The **Two's Complement Encoding** is the most common coding system for signed numbers so we will study it
- ▶ Alternatives exist
 - ▶ Signed magnitude: leading bit indicates pos (0) or neg (1)
 - ▶ One's complement: invert bits to go between positive negative
- ▶ Great advantage of two's complement: **signed and unsigned arithmetic are identical**
- ▶ Hardware folks only need to make one set of units for both unsigned and signed arithmetic

Summary of Two's Complement

TL;DR: *Most significant bit is a negative power of two.*

UNSIGNED BINARY

7654 3210 : position
ABCD EFGH : 8 bits
A: 0/1 * $+(2^7)$ *POS*
B: 0/1 * $+(2^6)$
C: 0/1 * $+(2^5)$
...
H: 0/1 * $+(2^0)$

TWO's COMPLEMENT (signed)

7654 3210 : position
ABCD EFGH : 8-bits
A: 0/1 * $-(2^7)$ *NEG*
B: 0/1 * $+(2^6)$
C: 0/1 * $+(2^5)$
...
H: 0/1 * $+(2^0)$

UNSIGNED BINARY

7654 3210 : position
1000 0000 = +128
1000 0001 = +129
1000 0011 = +131
1111 1111 = +255
0000 0000 = 0
0000 0001 = +1
0000 0101 = +5
0111 1111 = +127

TWO's COMPLEMENT (signed)

7654 3210 : position
1000 0000 = -128
1000 0001 = -127 = -128+1
1000 0011 = -125 = -128+1+2
1111 1111 = -1 = -128+1+2+4+...+64
0000 0000 = 0 [+127]
0000 0001 = +1
0000 0101 = +5
0111 1111 = +127

Two's Complement Notes

Unsigned/Signed Equivalents

Unsigned	1000 0110 = 134
Signed	1000 0110 = -121 = 134 - 256
Unsigned	1111 0001 = 241
Signed	1111 0001 = -15 = 241 - 256
Unsigned	0011 0011 = 51
Signed	0011 0011 = 51

When/Why X-256?

- ▶ Leading (leftmost) bit is 1
- ▶ Counted as 128 in Unsigned
- ▶ Counts as -128 in Signed
- ▶ Subtract 256 to compensate

Negation in Two's Complement

int y = -x;

- ▶ Unary Minus operator
- ▶ **Invert bits, Add 1**
- ▶ Works for both Pos→Neg and Neg→Pos

```
~ 0110 1000 +104 : negate
-----
  1001 0111 inverted
+           1
-----
  1001 1000 = -104
```

```
~ 1001 1000 = -104 : negate
-----
  0110 0111 = +103 inverted
+           1
-----
  0110 1000 = +104
```

Exercise: Two's Complement Conversions

Fill in the missing entries in the following table

Bits		Hex	Decimal Unsigned	Decimal Signed
0010 0011	A: _____		35	B: ____
1001 0110		0x96	C: ____	D: ____
1111 1111	E: _____		F: ____	G: ____
<i>OPTIONAL</i>				
H: _____		0x3E	I: ____	J: ____
K: _____	L: _____		M: ____	-35

Answers: Two's Complement Conversions

Bits		Hex	Decimal Unsigned	Decimal Signed
0010 0011	A:	0x23	35	B: 35
1001 0110		0x96	C: 150	D: -106
1111 1111	E:	0xFF	F: 255	G: -1
<i>OPTIONAL</i>				
H: 0011 1110		0x3E	I: 62	J: 62
K: 1101 1101	L:	0xDD	M: 221	-35

0010 0011 = 35 (Box B)

K / L / M: Converting 35 to
-35 decimal/bits can be done via
(-35+256) AND/OR via Invert
Bits + 1

~(0010 0011)

1101 1100

+ 1

1101 1101 (Box K)

Overflow

- ▶ Sums that exceed the representation of the bits associated with the integral type **overflow**
- ▶ Excess significant bits are **dropped**
- ▶ Addition can result in a sum smaller than the summands, even for two positive numbers (!?)
- ▶ Integer arithmetic in fixed bits is a mathematical **ring**

Examples of Overflow in 8 bits

ADDITION #3 OVERFLOW

1 1111 111 <-carries

1111 1111 = 255

+ 0000 0001 = 1

1 0000 0000 = 256

x drop 9th bit

0000 0000 = 0

ADDITION #4 OVERFLOW

1 1 <-carries

1010 1001 = 169

+ 1100 0001 = 193

1 0110 1010 = 362

x drop 9th bit

0110 1010 = 106

Underflow

- ▶ **Underflow** occurs in unsigned arithmetic when values go below 0 (no longer positive)
- ▶ Pretend that there is an extra significant bit to carry out subtraction
- ▶ Subtracting a positive integer from a positive integer may result in a **larger** positive integer (!?)
- ▶ Integer arithmetic in fixed bits is a mathematical **ring**

Examples of 8-bit Underflow

SUBTRACTIION #2 UNDERFLOW

?<-carries

0000 0000 = 0

- 0000 0001 = 1

VVVVVVVVVVVVVV

?<-carries

1 0000 0000 = 256 (pretend)

- 0000 0001 = 1

VVVVVVVVVVVVVV

x 2<-carries

0 1111 1110 = 256

- 0000 0001 = 1

1111 1111 = 255

Overflow and Underflow In C Programs

- ▶ See `over_under_flow.c` for demonstrations in a C program.
- ▶ **No runtime errors** for under/overflow
- ▶ Good for hashing and cryptography
- ▶ Bad for most other applications: system critical operations should use checks for overflow / underflow
- ▶ Textbook mentions the [Ariane Rocket Crash](#) which was due to overflow of an integer converted from a floating point value

The Ariane explosion is an instructive example for several reasons.
(1) Software re-use caused the problem subverting the usual wisdom of relying on tested software; hardware changes ALWAYS trump software.
(2) Sometimes computer science IS rocket science
- ▶ Assembly provides condition codes indicating when overflow occurs but checking in C is tricky and painful¹

¹Many compilers like GCC can generate assembly instructions that will detect overflow and abort programs. See the demo program `overflow_detect.c` and GCC's `-ftrapv` option.

Integer Ops and Speed

- ▶ Along with Addition and Subtraction, **Multiplication and Division** can also be done in binary
- ▶ Algorithms are the same as base 10 but more painful to do by hand
- ▶ This pain is reflected in hardware speed of these operations
- ▶ The **Arithmetic and Logic Unit (ALU)** does integer ops in the machine
- ▶ A **clock** ticks in the machine at some rate like 3Ghz (3 billion times per second)

- ▶ Under ideal circumstances, typical ALU Op speeds are

Operation	Cycles
Addition	1
Logical	1
Shifts	1
Subtraction	1
Multiplication	3
Division	>30

- ▶ Due to disparity, it is worth knowing about **relation** between multiply/divide and **bitwise** operations
- ▶ Compiler often uses such tricks: shift rather than multiply/divide

Mangling Bits Builds Muscle

Below illustrates difference between logical and bitwise operations.

```
int x1 = 12 || 10; // truthy (Logical OR)
int xb = 12 | 10;  // 14      (Bitwise OR)
int y1 = 12 && 10; // truthy (Logical AND)
int yb = 12 & 10;  // 8       (Bitwise AND)
int zb = 12 ^ 10;  // 6       (Bitwise XOR)
int w1 = !12;      // falsey  (Logical NOT)
int wb = ~12;      // 3       (Bitwise NOT/INVERT)
```

- ▶ Bitwise ops evaluate on a per-bit level
- ▶ 32 bits for int, 4 bits shown

Bitwise OR	Bitwise AND	Bitwise XOR	Bitwise NOT
1100 = 12	1100 = 12	1100 = 12	
1010 = 10	& 1010 = 10	^ 1010 = 10	~ 1100 = 12
-----	-----	-----	-----
1110 = 14	1000 = 8	0110 = 6	0011 = 3

Bitwise Shifts

- ▶ **Shift** operations move bits within a field of bits

- ▶ Shift operations are

```
x = y << k; // left shift y by k bits, store in x
```

```
x = y >> k; // right shift y by k bits, store in x
```

- ▶ All integral types can use shifts: long, int, short, char

- ▶ **Not applicable** to pointers or floating point

- ▶ Examples in 8 bits

```
//          76543210
```

```
char x = 0b00010111; // 23
```

```
char y = x << 2;      // left shift by 2
```

```
// y = 0b01011100; // 92
```

```
// x = 0b00010111; // not changed
```

```
char z = x >> 3;      // right shift by 3
```

```
// z = 0b00000010; // 2
```

```
// x = 0b00010111; // not changed
```

```
char n = 0b10000000; // -128, signed
```

```
char s = n >> 4;      // right shift by 4
```

```
// s = 0b11111000; // -8, sign extension
```

```
// right shift >> is "arithmetic"
```

Shifty Arithmetic Tricks

- ▶ Shifts with add/subtract can be used instead of multiplication and division
- ▶ Turn on optimization: `gcc -O3 code.c`
- ▶ Compiler automatically does this if it thinks it will save cycles
- ▶ *Sometimes* programmers should do this but better to convince compiler to do it for you, **comment** if doing manually

Multiplication

```
//          76543210
char x = 0b00001010; // 10
char x2 = x << 1;      // 10*2
// x2 = 0b00010100;    // 20
char x4 = x << 2;      // 10*4
// x4 = 0b00101000;    // 40
char x7 = (x << 3)-x;  // 10*7
// x7 = (x * 8)-x;      // 10*7
// x7 = 0b01000110;    // 70
//          76543210
```

Division

```
//          76543210
char y = 0b01101110;  // 110
char y2 = y >> 1;      // 110/2
// y2 = 0b00110111;    // 55
char y4 = y >> 2;      // 110/4
// y4 = 0b00011011;    // 27
char z = 0b10101100;  // -84
char z2 = z >> 2;      // -84/4
// z2 = 0b11101011;    // -21
// right shift sign extension
```

Exercise: Checking / Setting Bits

Use a combination of bit shift / bitwise logic operations to...

1. Check if bit *i* of `int x` is set (has value 1)
2. Clear bit *i* (set bit at index *i* to value 0)

Show C code for this

```
{
    int x = ...;
    int i = ...;
    if( ??? ) { // ith bit of x is set
        printf("set!\n");
    }

    i = ...;
    ???;
    printf("ith bit of x now cleared to 0\n");
}
```

Answers: Checking / Setting Bits

1. Check if bit *i* of int *x* is set (has value 1)

```
int x = ...;
int mask = 1; // or 0b0001 or 0x01 ...
int shifted = mask << i; // shifted  0b00...010..00
if(x & shifted){          //          x & 0b10...010..01
    ...                  //          -----
}                        //          0b00...010..00
```

2. Clear bit *i* (set bit at index *i* to value 0)

```
int x = ...;
int mask = 1; // or 0b0001 or 0x01 ...
int shifted = mask << i; // shifted  0b00...010..00
int inverted = ~shifted; // inverted 0b11...101..11
x = x & inverted;        //          x & 0b10...010..01
...                      //          -----
                        //          0b10...000..01
```

Showing Bits

- ▶ `printf()` capabilities:
 - `%d` as Decimal
 - `%x` as Hexadecimal
 - `%o` as Octal
 - `%c` as Character
- ▶ No specifier for binary
- ▶ Can construct such with bitwise operations
- ▶ Code pack contains two codes to do this
 - ▶ `printbits.c`: single args printed as 32 bits
 - ▶ `showbits.c`: multiple args printed in binary, hex, decimal

- ▶ Showing bits usually involves shifting and bitwise AND &

- ▶ Example from `showbits.c`

```
#define INT_BITS 32

// print bits for x to screen
void showbits(int x){
    for(int i=INT_BITS-1; i>=0; i--){
        int mask = 1 << i;
        if(mask & x){
            printf("1");
        } else {
            printf("0");
        }
    }
}
```

Bit Masking

- ▶ Semi-common for functions to accept bit patterns which indicate true/false options
- ▶ Frequently makes use of bit **masks** which are constants associated with specific bits
- ▶ Example: Unix permissions might be...

```
#define S_IRUSR 0b100000000 // User   Read
#define S_IWUSR 0b010000000 // User   Write
#define S_IXUSR 0b001000000 // User   Execute
#define S_IRGRP 0b000100000 // Group  Read
...
#define S_IWOTH 0b000000010 // Others Write
#define S_IXOTH 0b000000001 // Others Execute
```

- ▶ Use them to create options to C functions like

```
int permissions = S_IRUSR|S_IWUSR|S_RGRP;
chmod("/home/kauffman/solution.zip",permissions);
```

Unix Permissions with Octal

- ▶ Octal arises associated with **Unix file permissions**
- ▶ Every file has 3 permissions for 3 entities
- ▶ Permissions are true/false so a single bit will suffice
- ▶ `ls -l`: long list files, shows permissions
- ▶ `chmod 665 somefile.txt`:
change permissions of
somefile.txt to those
shown to the right
 - binary octal
 - 110110101 = 665
 - rw-rw-r-x somefile.txt
 - U G O
 - S R T
 - E O H
 - R U E
 - P R
- ▶ `chmod 777 x.txt`: read /
write / exec for everyone
- ▶ `chmod` also honors letter
versions like r and w
- ▶ `chmod u+x script.sh #`
make file executable

Readable chmod version:
`chmod u=rw,g=rw,o=rx somefile.txt`

Endianness: Byte ordering in Memory

- ▶ Single bytes like ASCII characters lay out sequentially in memory in increasing address
- ▶ Multi-byte entities like 4-byte ints require decisions on byte ordering
- ▶ We think of a 32-bit int like this

	Most Significant	<----->				Least Significant			
Binary:	0000	0000	0000	0000	0001	1000	1110	1001	
	0	0	0	0	1	8	E	9	
Hex :	000018E9								
Decimal:	6377								

- ▶ There are 2 Options to for ordering multi-byte data in memory
 - ▶ **Little Endian:** Least Significant byte at low address
 - ▶ **Big Endian:** Most Significant Byte at low address
- ▶ Example: Integer starts at address #1024

	Address							
LittleEnd:	#1027	#1026	#1025	#1024				
Binary:	0000	0000	0000	0000	0001	1000	1110	1001
	0	0	0	0	1	8	E	9
BigEnd:	#1024	#1025	#1026	#1027				
	Address							

Little Endian vs. Big Endian

- ▶ Most modern machines use **Little Endian** ordering by default
- ▶ Some processor (ARM) support both Little / Big Endian BUT one is chosen at startup and used until turned off
- ▶ Both Big and Little Endian have (minor) engineering trade-offs
- ▶ At one time debated hotly among hardware folks: a la *Gulliver's Travels* conflicts
- ▶ **Intel Chips** were little endian and have dominated computing for several decades, set the precedent for modern platforms
- ▶ Big endian byte order shows up in **network programming**: sending bytes over the network is done in big endian ordering
- ▶ **Examine** `show_endianness.c` : uses C code to print bytes in order, reveals whether a machine is Little or Big Endian

Output of show_endianness.c

```
1 // show_endianness.c: Shows endiannes layout of a binary number in
2 // memory. Intel machines and some ARM machines (Apple M1) are little
3 // endian so bytes will print least significant earlier.
4 #include <stdio.h>
5
6 int main(){
7     int bin = 0b0000000000000000000000001100011101001;    // 6377
8     //           |   |   |   |   |   |   |   |
9     //           0   0   0   0   1   8   e   9
10    printf("%d\n%08x\n",bin,bin);                          // show decimal and hex representation of b
11    char *ptr = (char *) &bin;                               // pointer to beginning of bin
12    for(int i=0; i<4; i++){                                  // print bytes of bin from low to high
13        printf("%hhx ", ptr[i]);                             // memory address
14    }                                                         // '%hhx' : 1-byte char in hex
15    printf("\n");                                             // '%hx'  : 2-byte short in hex
16    return 0;                                                 // '%x'   : 4-byte int in hex
17 }
```

```
>> gcc show_endianness.c
>> ./a.out
6377
000018e9
e9 18 0 0
```

Notice: num prints with value 18e9 but bytes appear in reverse order e9 18 when run on a Little Endian machine: the “littlest” byte appears earliest in memory

Floating Point Encoding

- ▶ Floating point numbers like `float` / `double` use a completely different encoding scheme from integers
- ▶ This is covered in an optional Video lecture and associated slides
- ▶ NOT part of the main line course: will not appear in a major way on projects or exams
- ▶ Those who are interested are free to peruse the video, slides, and associated code

Students are responsible for knowing that the encoding of integers is different from that of floating point numbers.

- ▶ A 32-bit pattern can be interpreted as encoding an Integer between -2,147,483,648 and 2,147,483,647
OR...
- ▶ The same 32 bits can be interpreted DIFFERENTLY as encoding a Real number in the range $\pm 3.40282347\text{E}\pm 38$